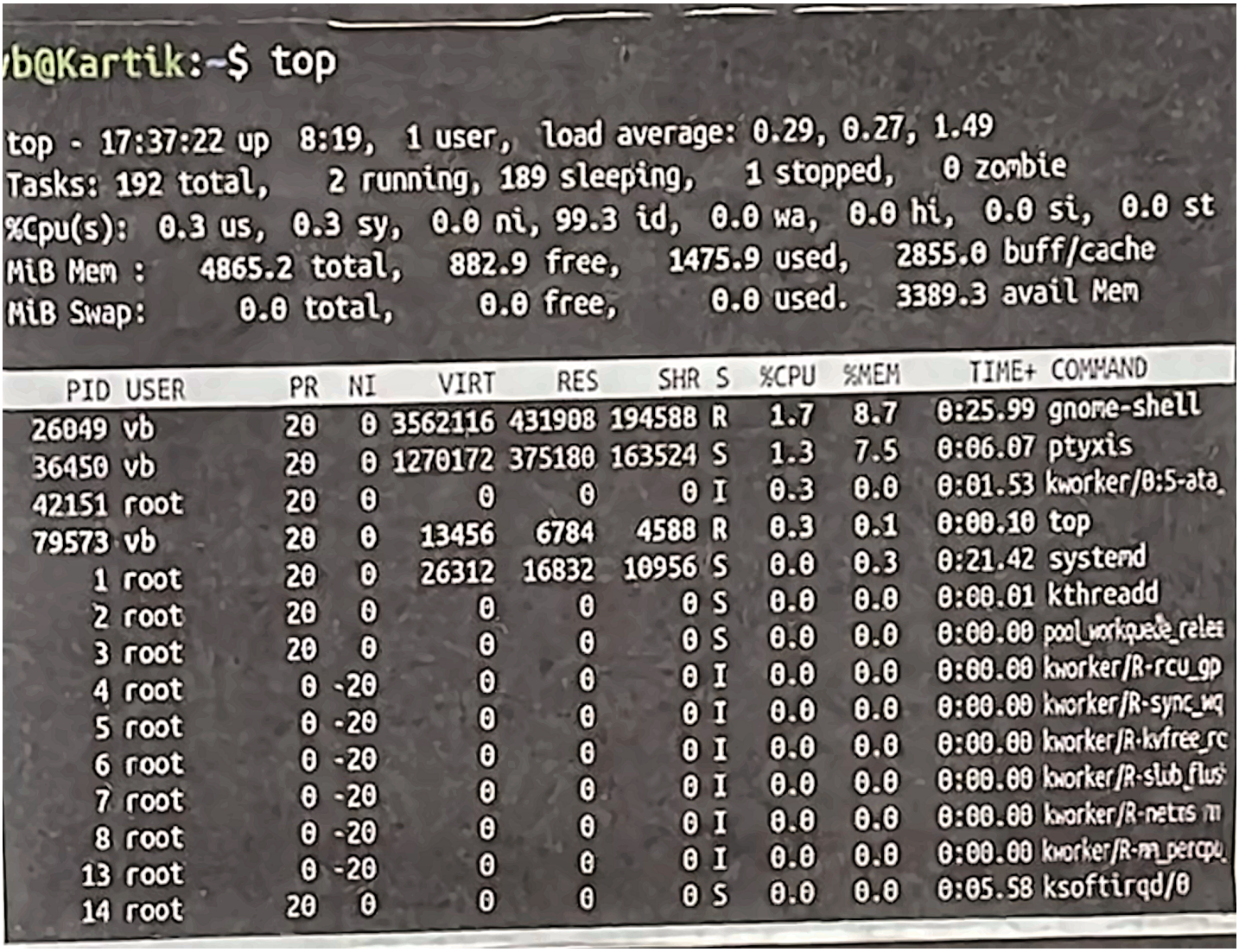
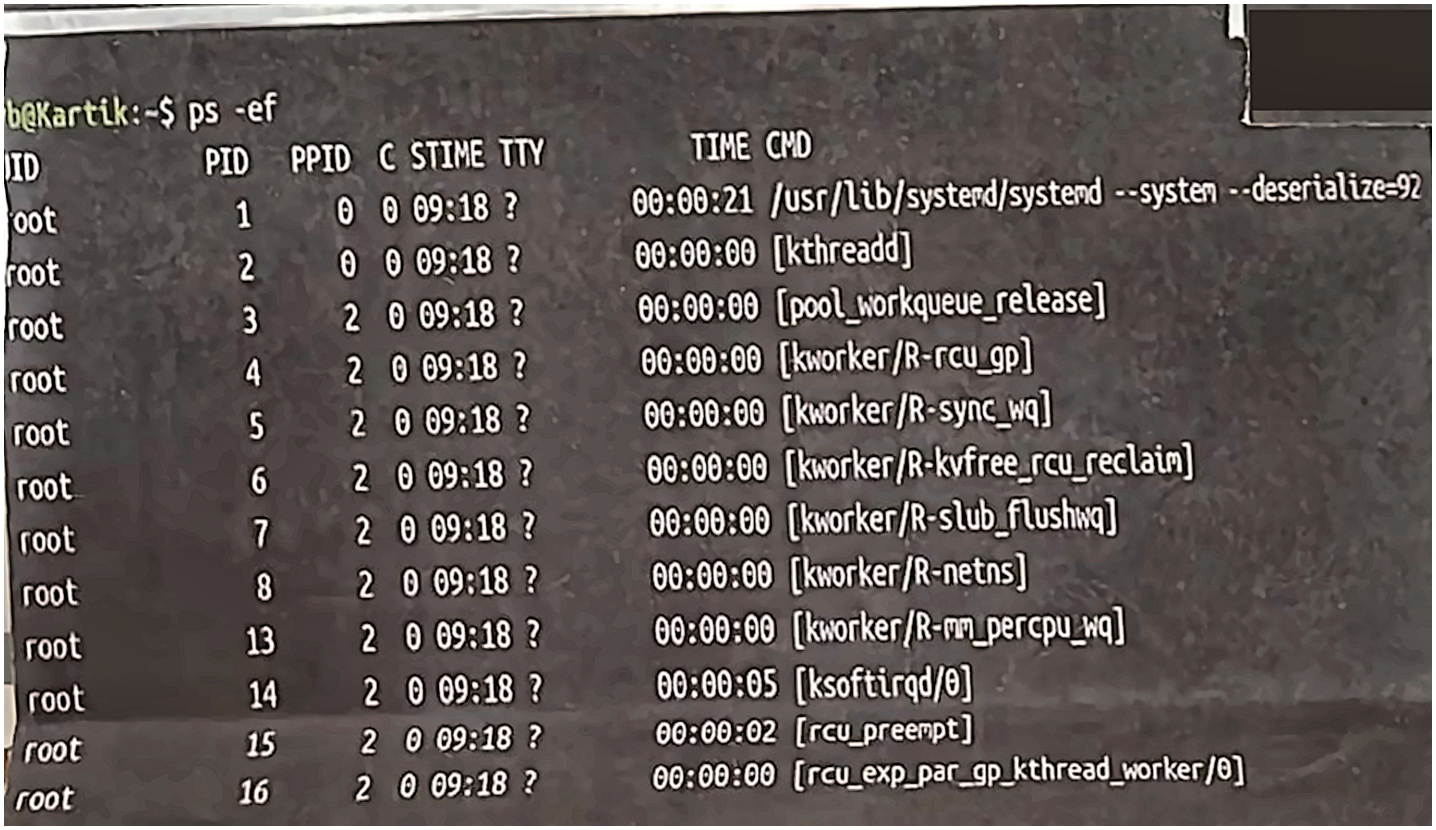
```
vb@Kartik:~$ jobs
[2]+  Stopped                 sleep 200
[3]   Running                 sleep 500 &
[5]-  Running                 sleep 600 &
vb@Kartik:~$ ps
    PID TTY          TIME CMD
  36484 pts/0    00:00:00 bash
  77631 pts/0    00:00:00 sleep
  79482 pts/0    00:00:00 sleep
  79494 pts/0    00:00:00 sleep
  79541 pts/0    00:00:00 ps
vb@Kartik:~$ kill -9 79482
[3]   Killed                  sleep 500
vb@Kartik:~$ kill -15 77631
vb@Kartik:~$ kill -15 79494\
>
[5]-  Terminated              sleep 600
vb@Kartik:~$ ps -o pid,ni,cmd
    PID  NI CMD
  36484   0 /usr/bin/bash
  77631   0 sleep 200
  79554   0 ps -o pid,ni,cmd
vb@Kartik:~$ renice -4 -p 77631
renice: failed to set priority for 77631 (process ID): Permission denied
vb@Kartik:~$ sudo renice 5 -p 77631
[sudo: authenticate] Password:
77631 (process ID) old priority 0, new priority 5
```

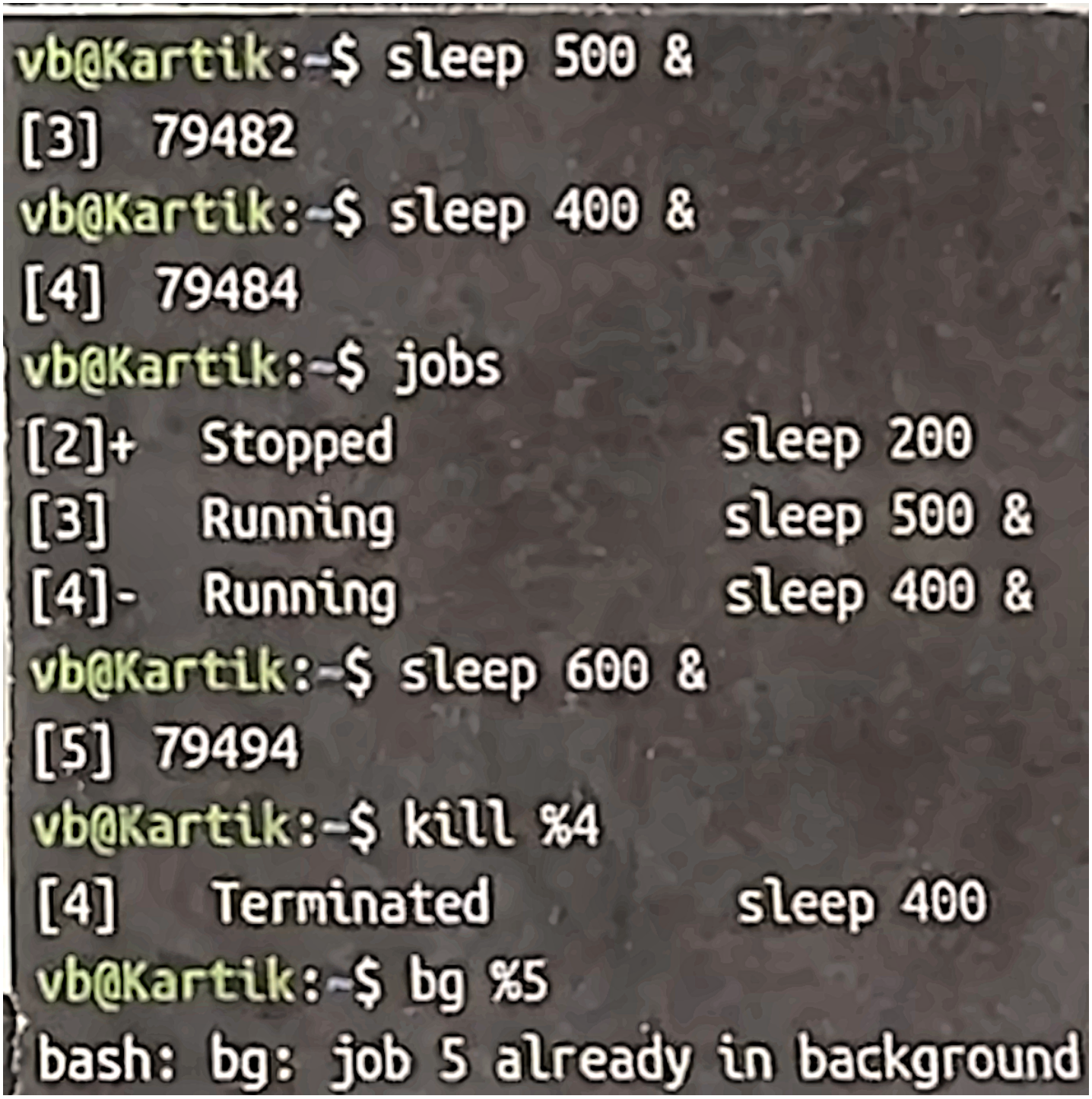
vb@Kartik:~$ top


vb@Kartik:~$ top
top - 17:37:22 up 8:19, 1 user, load average: 0.29, 0.27, 1.49
Tasks: 192 total, 2 running, 189 sleeping, 1 stopped, 0 zombie
%Cpu(s): 0.3 us, 0.3 sy, 0.0 ni, 99.3 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 4865.2 total, 882.9 free, 1475.9 used, 2855.0 buff/cache
MiB Swap: 0.0 total, 0.0 free, 0.0 used. 3389.3 avail Mem
PID USER PR NI VIRT RES SHR S %CPU %MEM TIME+ COMMAND
26049 vb 20 0 3562116 431908 194588 R 1.7 8.7 0:25.99 gnome-shell
36450 vb 20 0 1270172 375180 163524 S 1.3 7.5 0:06.07 ptyxis
42151 root 20 0 0 0 0 I 0.3 0.0 0:01.53 kworker/0:5-ata
79573 vb 20 0 13456 6784 4588 R 0.3 0.1 0:00.10 top
1 root 20 0 26312 16832 10956 S 0.0 0.3 0:21.42 systemd
2 root 20 0 0 0 0 S 0.0 0.0 0:00.01 kthreadd
3 root 20 0 0 0 0 S 0.0 0.0 0:00.00 pool_workqueue_rele
4 root 0 -20 0 0 0 I 0.0 0.0 0:00.00 kworker/R-rcu_gp
5 root 0 -20 0 0 0 I 0.0 0.0 0:00.00 kworker/R-sync_wq
6 root 0 -20 0 0 0 I 0.0 0.0 0:00.00 kworker/R-kvfree_rc
7 root 0 -20 0 0 0 I 0.0 0.0 0:00.00 kworker/R-slub_flus
8 root 0 -20 0 0 0 I 0.0 0.0 0:00.00 kworker/R-netns
13 root 0 -20 0 0 0 I 0.0 0.0 0:00.00 kworker/R-mm_percpu
14 root 20 0 0 0 0 S 0.0 0.0 0:05.58 ksoftirqd/0

```
vb@Kartik:~$ ps
    PID TTY          TIME CMD
  36484 pts/0    00:00:00 bash
  77631 pts/0    00:00:00 sleep
  79482 pts/0    00:00:00 sleep
  79494 pts/0    00:00:00 sleep
  79512 pts/0    00:00:00 ps
vb@Kartik:~$ ps -e
    PID TTY          TIME CMD
      1 ?        00:00:21 systemd
      2 ?        00:00:00 kthreadd
      3 ?        00:00:00 pool_workqueue_release
      4 ?        00:00:00 kworker/R-rcu_gp
      5 ?        00:00:00 kworker/R-sync_wq
      6 ?        00:00:00 kworker/R-kvfree_rcu_recla
      7 ?        00:00:00 kworker/R-slub_flushwq
      8 ?        00:00:00 kworker/R-netns
     13 ?        00:00:00 kworker/R-mm_percpu_wq
     14 ?        00:00:05 ksoftirqd/0
     15 ?        00:00:02 rcu_preempt
     16 ?        00:00:00 rcu_exp_par_gp_kthread_worker
     17 ?        00:00:00 rcu_exp_gp_kthread_worker
     18 ?        00:00:00 migration/0
     19 ?        00:00:00 kprobe-optimizer
     20 ?        00:00:00 idle_inject/0
     21 ?        00:00:00 cpuhp/0
```



```
b@Kartik:~$ ps -ef
UID          PID    PPID  C STIME TTY          TIME CMD
root           1       0  0 09:18 ?        00:00:21 /usr/lib/systemd/systemd --system --deserialize=92
root           2       0  0 09:18 ?        00:00:00 [kthreadd]
root           3       2  0 09:18 ?        00:00:00 [pool_workqueue_release]
root           4       2  0 09:18 ?        00:00:00 [kworker/R-rcu_gp]
root           5       2  0 09:18 ?        00:00:00 [kworker/R-sync_wq]
root           6       2  0 09:18 ?        00:00:00 [kworker/R-kvfree_rcu_reclaim]
root           7       2  0 09:18 ?        00:00:00 [kworker/R-slub_flushwq]
root           8       2  0 09:18 ?        00:00:00 [kworker/R-netns]
root          13       2  0 09:18 ?        00:00:00 [kworker/R-mm_percpu_wq]
root          14       2  0 09:18 ?        00:00:05 [ksoftirqd/0]
root          15       2  0 09:18 ?        00:00:02 [rcu_preempt]
root          16       2  0 09:18 ?        00:00:00 [rcu_exp_par_gp_kthread_worker/0]
```


vb@Kartik:~$ sleep 500 &
[3] 79482
vb@Kartik:~$ sleep 400 &
[4] 79484
vb@Kartik:~$ jobs
[2]+ Stopped sleep 200
[3] Running sleep 500 &
[4]- Running sleep 400 &
vb@Kartik:~$ sleep 600 &
[5] 79494
vb@Kartik:~$ kill %4
[4] Terminated sleep 400
vb@Kartik:~$ bg %5
bash: bg: job 5 already in background